

1. To write a Python program to add to numbers entered by the user.

```
1 num1 = float(input("Enter the first number: "))
2
3 num2 = float(input("Enter the second number: "))
4
5 sum = num1 + num2
6
7 print(f"The sum of {num1} and {num2} is: {sum}")
```

Enter the first number: 2
Enter the second number: 6
The sum of 2.0 and 6.0 is: 8.0

2. To write a Python program to interchange the values of two variables.

```
1 num1 = float(input("Enter the first number: "))
2 num2 = float(input("Enter the second number: "))
3 temp = num1
4 num1 = num2
5 num2 = temp
6 print(f"After interchange The value of first number and second number are: {num1} and {num2}")
```

Enter the first number: 7
Enter the second number: 6
After interchange The value of first number and second number are: 6.0 and 7.0

3. To write a Python program to find the largest among three numbers using conditional statements.

```
1 num1 = float(input("Enter the first number: "))
2 num2 = float(input("Enter the second number: "))
3 if(num1 > num2):
4     print(f"{num1} is greater than {num2}")
5 else:
6     print(f"{num2} is greater than {num1}")
```

Enter the first number: 5
Enter the second number: 45786668
45786668.0 is greater than 5.0

4. To write a Python program to find the largest among three numbers using conditional statements.

```
1 num1 = float(input("Enter the first number: "))
2 num2 = float(input("Enter the second number: "))
3 num3 = float(input("Enter the third number: "))
4 if(num1 > num2 and num1 > num3):
5     print(f"{num1} is greater than {num2} and {num3}")
6 elif(num2 > num1 and num2 > num3):
7     print(f"{num2} is greater than {num1} and {num3}")
8 else:
9     print(f"{num3} is greater than {num1} and {num2}")
```

Enter the first number: 7
Enter the second number: 9
Enter the third number: 456
456.0 is greater than 7.0 and 9.0

5. To write a Python program to determine whether a number is positive, negative, or zero

```
1 num = float(input("Enter the number: "))
2 if(num < 0):
3     print(f"{num} is negative number")
4 elif(num > 0):
5     print(f"{num} is positive number")
6 else:
7     print(f"{num} is zero")
8
```

Enter the number: -97
-97.0 is negative number

6. To write a Python program to check whether a given year is a leap year.

```
1 year = int(input("Enter the year: "))
```

```

1 year = int(input("Enter the year: "))
2 if(year % 4 == 0 and year % 100 != 0 or year % 400 == 0):
3     print(f"{year} is a leap year")
4 else:
5     print(f"{year} is not a leap year")

```

Enter the year: 2025
2025 is not a leap year

7. To write a Python program to perform basic arithmetic operations using operators.

```

1 num1 = float(input("Enter the first number: "))
2 num2 = float(input("Enter the second number: "))
3 print(f"The sum of {num1} and {num2} is: {num1 + num2}")
4 print(f"The difference of {num1} and {num2} is: {num1 - num2}")
5 print(f"The product of {num1} and {num2} is: {num1 * num2}")
6 print(f"The division of {num1} and {num2} is: {num1 / num2}")
7 print(f"The modulus of {num1} and {num2} is: {num1 % num2}")

```

Enter the first number: 3.89
Enter the second number: 4.76
The sum of 3.89 and 4.76 is: 8.65
The difference of 3.89 and 4.76 is: -0.8699999999999997
The product of 3.89 and 4.76 is: 18.5164
The division of 3.89 and 4.76 is: 0.8172268907563026
The modulus of 3.89 and 4.76 is: 3.89

```

1 num1 = int(input("Enter the first number: "))
2 num2 = int(input("Enter the second number: "))
3 print(f"The sum of {num1} and {num2} is: {num1 + num2}")
4 print(f"The difference of {num1} and {num2} is: {num1 - num2}")
5 print(f"The product of {num1} and {num2} is: {num1 * num2}")
6 print(f"The division of {num1} and {num2} is: {num1 / num2}")
7 print(f"The modulus of {num1} and {num2} is: {num1 % num2}")

```

Enter the first number: 3
Enter the second number: 6
The sum of 3 and 6 is: 9
The difference of 3 and 6 is: -3
The product of 3 and 6 is: 18
The division of 3 and 6 is: 0.5
The modulus of 3 and 6 is: 3

8. To write a Python program to display the multiplication table of a given number.

```

1 num = int(input("Enter the number: "))
2 for i in range(1, 11):
3     print(f"{num} x {i} = {num * i}")

```

Enter the number: 5
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
5 x 6 = 30
5 x 7 = 35
5 x 8 = 40
5 x 9 = 45
5 x 10 = 50

9. To write a Python program to calculate the factorial of a number using loop.

```

1 num = int(input("Enter a non-negative integer: "))
2
3 if num < 0:
4     print("Factorial does not exist for negative numbers")
5 elif num == 0:
6     print("The factorial of 0 is 1")
7 else:
8     factorial = 1
9     for i in range(1, num + 1):
10        factorial = factorial * i
11    print(f"The factorial of {num}! is {factorial}")

```

Enter a non-negative integer: 5
The factorial of 5! is 120

10. To write a Python program to find the sum of the first N natural numbers.

```
1 num = int(input("Enter a number: "))
2
3 if num < 1:
4     print("Please enter a positive integer.")
5 else:
6     sum = num * (num + 1) // 2
7     print(f"The sum of the first {num} natural numbers is: {sum}")
```

Enter a number: 6
The sum of the first 6 natural numbers is: 21

11. To write a Python program to print numbers from 1 to N using a for loop.

```
1 num = int(input("Enter a number: "))
2 if num < 1:
3     print("Please enter a positive integer.")
4 else:
5     for i in range(1, num + 1):
6         print(f"{i}")
```

Enter a number: 8
1
2
3
4
5
6
7
8

12. To write a Python program to print numbers in reverse order from N to 1.

```
1 num = int(input("Enter a number: "))
2 if num < 1:
3     print("Please enter a positive integer.")
4 else:
5     for i in range(num, 0, -1):
6         print(f"{i}")
```

Enter a number: 10
10
9
8
7
6
5
4
3
2
1

13. To write a Python program to print all even numbers up to a given limit.

```
1 num = int(input("Enter a number: "))
2 for i in range(1, num + 1):
3     if (i % 2 == 0):
4         print(f"{i}")
```

Enter a number: 10
2
4
6
8
10

14. To write a Python program to reverse the digits of a given number

```
1 num = int(input("Enter a number to reverse: "))
2
3 reversed_num = 0
4 original_num = num
5 temp_num = num
6
7 while temp_num > 0:
8     digit = temp_num % 10
9     reversed_num = reversed_num * 10 + digit
10    temp_num //= 10
11 print(f"The Original number is: {original_num}")
12 print(f"The reversed number is: {reversed_num}")
```

```
Enter a number to reverse: 342008
The Original number is: 342008
The reversed number is: 800243
```

15. To write a Python program to count the number of digits in a given number.

```
1 num = int(input("Enter a number: "))
2
3 num_str = str(num)
4 count = len(num_str)
5
6 print(f"The number of digits in {num} is: {count}")
```

```
Enter a number: 342008
The number of digits in 342008 is: 6
```

16. To write a Python program to calculate the sum of digit of a given number.

```
1 num = int(input("Enter a number: "))
2
3 sum_of_digits = 0
4 temp_num = abs(num)
5 while temp_num > 0:
6     digit = temp_num % 10
7     sum_of_digits += digit
8     temp_num //= 10
9
10 print(f"The sum of the digits of {num} is: {sum_of_digits}")
```

```
Enter a number: 342008
The sum of the digits of 342008 is: 17
```

17. To write a Python program to check whether a given number is prime.

```
1 num = int(input("Enter a number: "))
2
3 if num <= 1:
4     print(f"{num} is not a prime number.")
5 elif num == 2:
6     print(f"{num} is a prime number.")
7 else:
8     is_prime = True
9     for i in range(2, int(num**0.5) + 1):
10         if num % i == 0:
11             is_prime = False
12             break
13     if is_prime:
14         print(f"{num} is a prime number.")
15     else:
16         print(f"{num} is not a prime number.")
```

```
Enter a number: 5
5 is a prime number.
```

18. To write a Python program to generate the Fibonacci series up to N terms.

```
1 n_terms = int(input("Enter the number of terms for Fibonacci series: "))
2
3 a, b = 0, 1
4 count = 0
5
6 if n_terms <= 0:
7     print("Please enter a positive integer.")
8 elif n_terms == 1:
9     print("Fibonacci series up to 1 term:")
10    print(a)
11 else:
12    print(f"Fibonacci series up to {n_terms} terms:")
13    while count < n_terms:
14        print(a)
15        nth = a + b
16        a = b
17        b = nth
18        count += 1
```

```
Enter the number of terms for Fibonacci series: 5
Fibonacci series up to 5 terms:
0
```

```
1
1
2
3
```

19. To write a Python program to check whether a string is a palindrome.

```
1 s = input("Enter a string: ")
2
3 if s == s[::-1]:
4     print("Palindrome")
5 else:
6     print("Not a Palindrome")
```

```
Enter a string: Samiksha
Not a Palindrome
```

20. To write a Python program to check whether a given number is an Armstrong number.

```
1 num = int(input("Enter a number: "))
2
3 sum = 0
4 temp = num
5
6 while temp > 0:
7     digit = temp % 10
8     sum = sum + digit ** 3
9     temp = temp // 10
10
11 if sum == num:
12     print("Armstrong Number")
13 else:
14     print("Not an Armstrong Number")
```

```
Enter a number: 153
Armstrong Number
```